Track A – Research-Oriented Project

Milestone 3 – Final Research Report

Course: CSC-361 Machine Learning	**Semester:** Spring 2025
Track: Track A – Research-Oriented Project	**Due Date:** 30 May 2025
Instructor: Dr. Shafiq Ur Rehman Khan	**Max Marks:** 9 / 20
Mapped CLO: CLO-3	

Mapped Course Learning Outcome (CLO)

• CLO-3: Analyze machine learning model performance using appropriate preprocessing techniques and evaluation metrics.

Milestone Objective

Students will deliver a complete research paper documenting their full ML pipeline including model comparison, hyperparameter tuning, error analysis, and conclusions. This milestone evaluates the rigor of performance analysis using appropriate preprocessing techniques and evaluation metrics.

Required Deliverables

- Final Written Report (PDF, 6–10 pages in research-paper format)
- Presentation with viva (10–12 minutes + Q&A;)
- Clean and reproducible GitHub repository

Submission Requirements

- Abstract, Introduction, Related Work, Methodology, Experiments, Results, Discussion, Conclusion
- Comparison of at least two ML models/algorithms
- Hyperparameter tuning with documented results
- Confusion matrix and performance graphs
- Error analysis with discussion of failure cases
- Conclusion and future work

Evaluation Rubric — Total Marks: 9 / 20

Semester: Spring 2025 **Max Marks: 9**

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Instructor: Dr. Shafiq Ur Rehman Khan

Group Name / No.: **Submission Date:**

Criterion	Excellent (Full Marks)	Good (75%)	Satisfactory (50%)	Needs Work (25%)	Marks	CLO
Model Comparison (≥2 models)	Two+ models compared; results tabulated and thoroughly discussed	Two models compared; limited discussion	One model with brief mention of another	No comparison performed	2.0	CLO-3
Hyperparameter Tuning	Systematic tuning performed; best parameters documented with impact analysis	Tuning attempted; partial documentation	Minor tuning attempted; no documentation	No tuning performed	1.5	CLO-3
Evaluation & Performance Analysis	Comprehensive metrics, confusion matrix, and performance graphs with deep analysis	Metrics and graphs present with moderate analysis	Some metrics reported; limited visual analysis	Minimal or no evaluation analysis	2.0	CLO-3
Error Analysis	Thorough error analysis with identified failure cases and root cause discussion	Error analysis present but shallow	Brief mention of errors without analysis	No error analysis	1.0	CLO-3
Report Quality (Research Paper Format)	Complete research paper with all required sections; professional writing and citations	Most sections present; minor writing issues	Some sections missing; informal writing	Incomplete or disorganized report	1.0	CLO-3
Code Reproducibility	Repository fully organized; code runs end-to-end; results reproducible	Code mostly runnable; minor issues	Code partially functional	Code missing or non-functional	0.75	CLO-3
Presentation & Viva	Excellent delivery; all members answer viva questions confidently	Good delivery; most viva questions answered	Partial delivery; struggled with viva	Poor delivery; unable to answer questions	0.75	CLO-3

Marks Distribution Summary

Criterion	Max Marks	CLO
Model Comparison (≥2 models)	2.0	CLO-3
Hyperparameter Tuning	1.5	CLO-3

Evaluation & Performance Analysis	2.0	CLO-3
Error Analysis	1.0	CLO-3
Report Quality (Research Paper Format)	1.0	CLO-3
Code Reproducibility	0.75	CLO-3
Presentation & Viva	0.75	CLO-3
Total	**9.00**	

Important Notes: All group members must be able to explain their individual contribution during viva. Plagiarism in report or code will result in zero marks for the entire group. Accuracy alone is not an acceptable evaluation metric — use multiple metrics appropriate to the ML task.